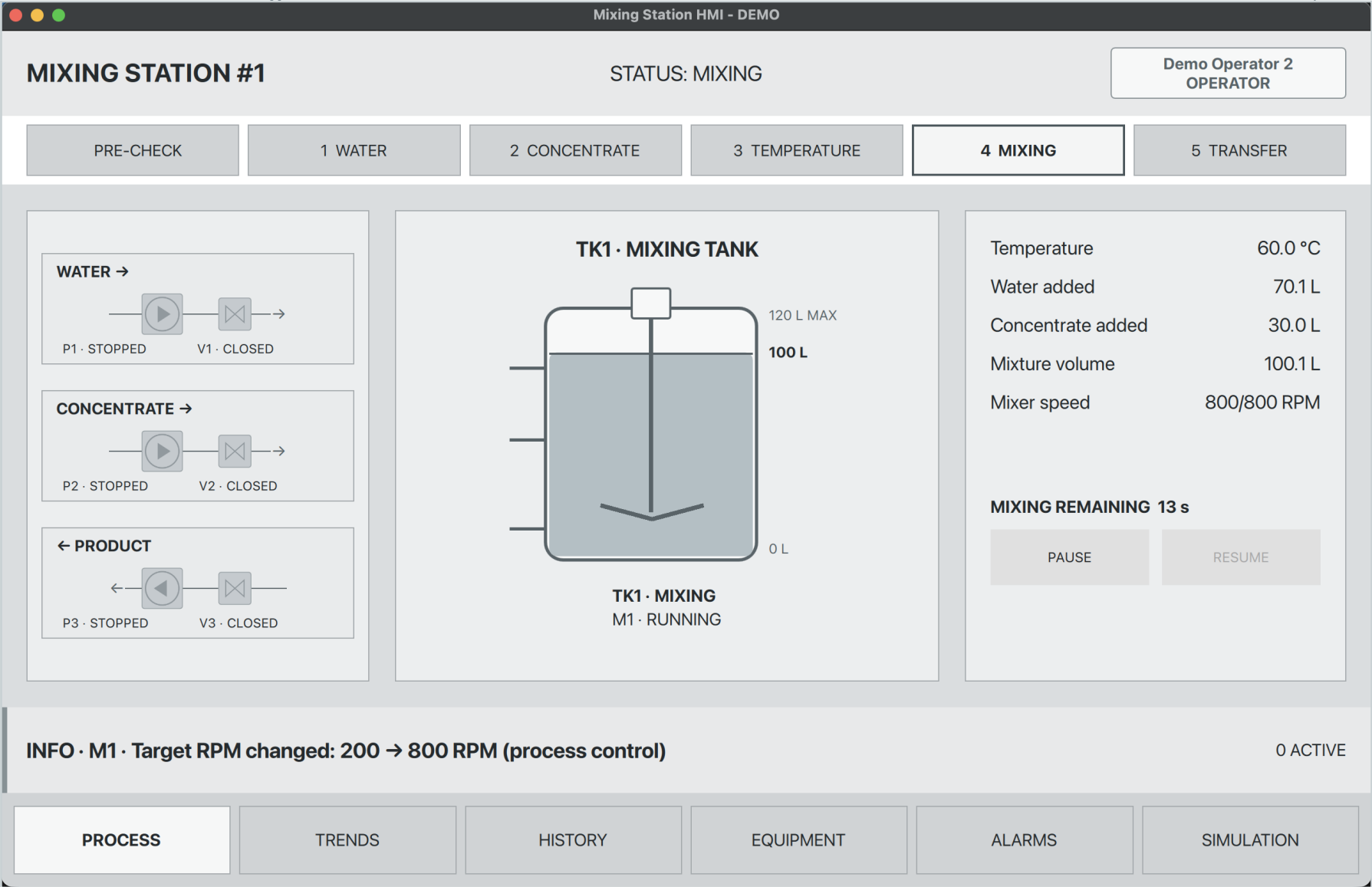


MixingStationHMI

A touch-oriented batch mixing station simulation

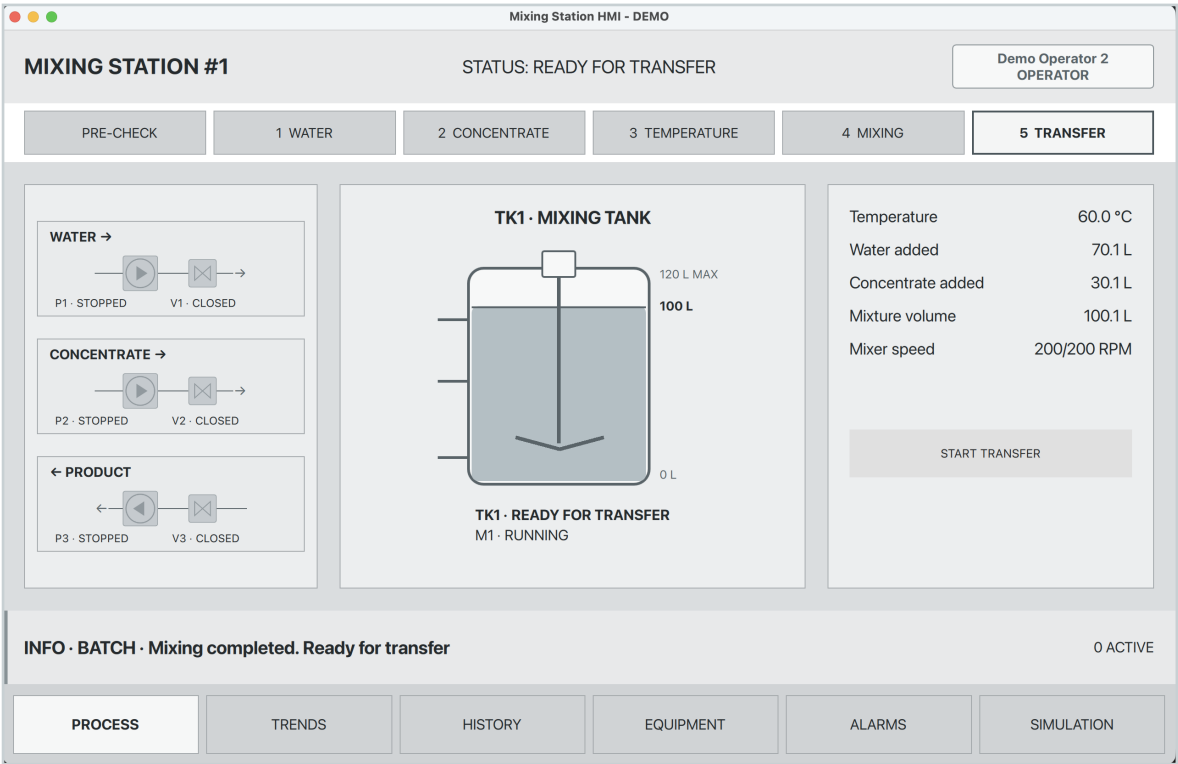
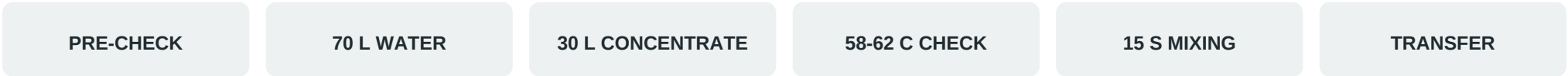
APPLICATION GUIDE | VERSION 0.1



Process view during the 15-second mixing stage

The batch sequence

A 100 L target batch runs through explicit process states and requires an operator to start transfer.



Pre-check confirms the required starting conditions, including a connected mixer and an operator selection.

Water and concentrate are dosed in separate stages. Process values show the amounts added and the current mixture volume.

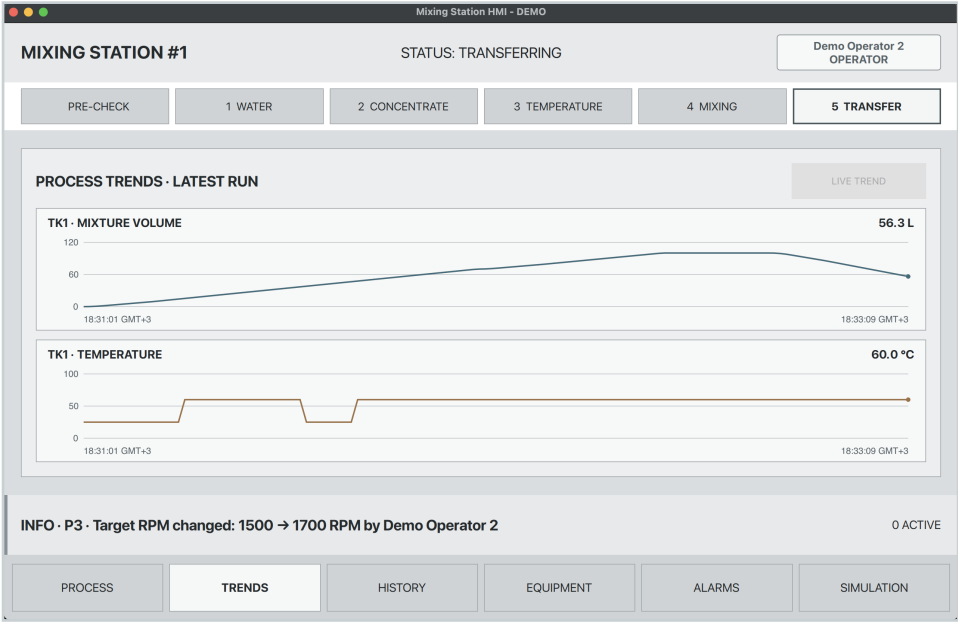
Temperature must be in the 58-62 C range before mixing. During mixing, the M1 target is 800 RPM; it returns to 200 RPM while waiting for transfer.

Transfer starts only when the operator requests it. The process can then be prepared for another batch.

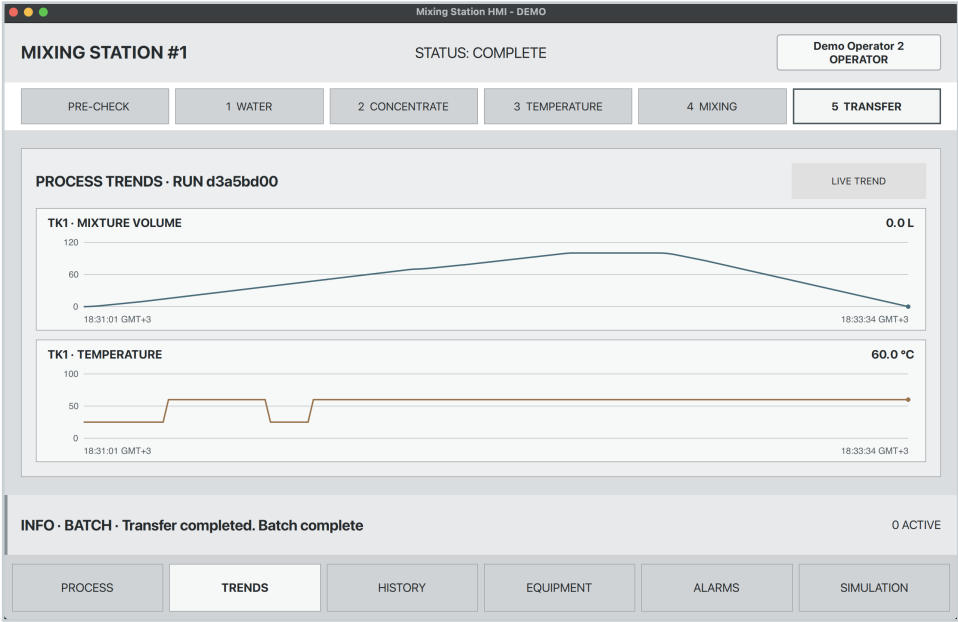
Ready for transfer: M1 keeps running at 200 RPM until the operator starts P3/V3.

Trends and run history

Live process data is visible during a run; completed runs retain events and trend samples in SQLite.

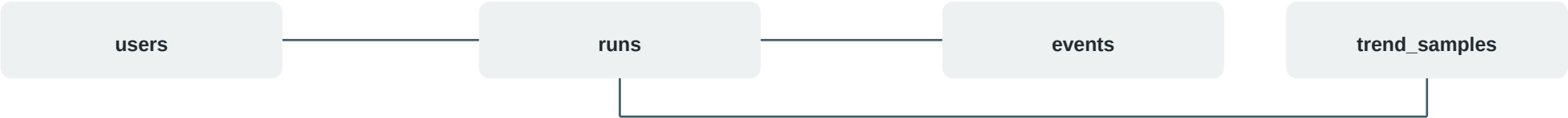


Live trends: volume and temperature while the process runs.



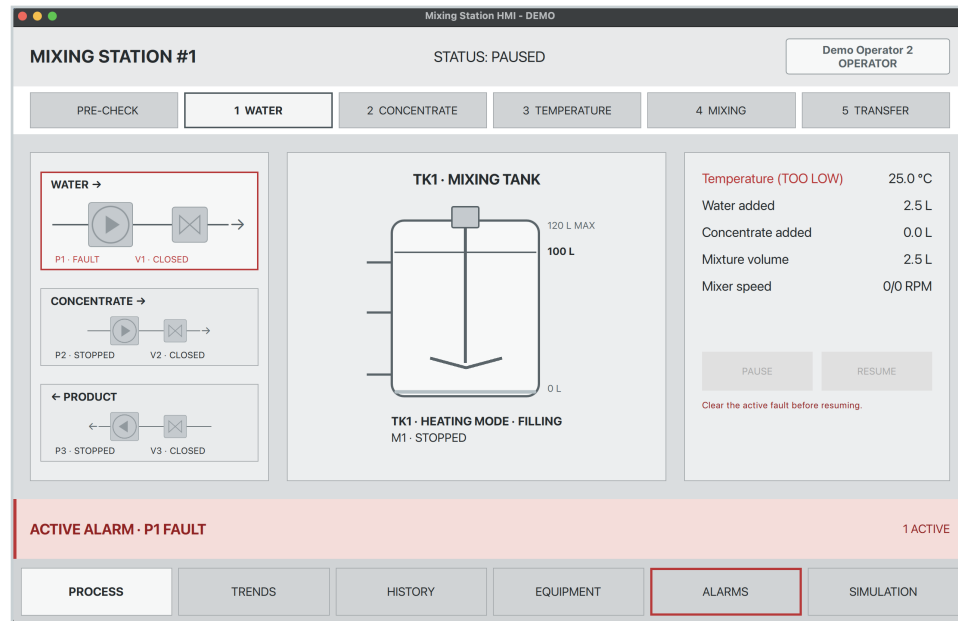
Historical trends: saved samples from the selected run.

The History tab lists runs and their events. Selecting a run opens its saved trend. Run IDs connect each batch or drain operation with its events and trend samples.

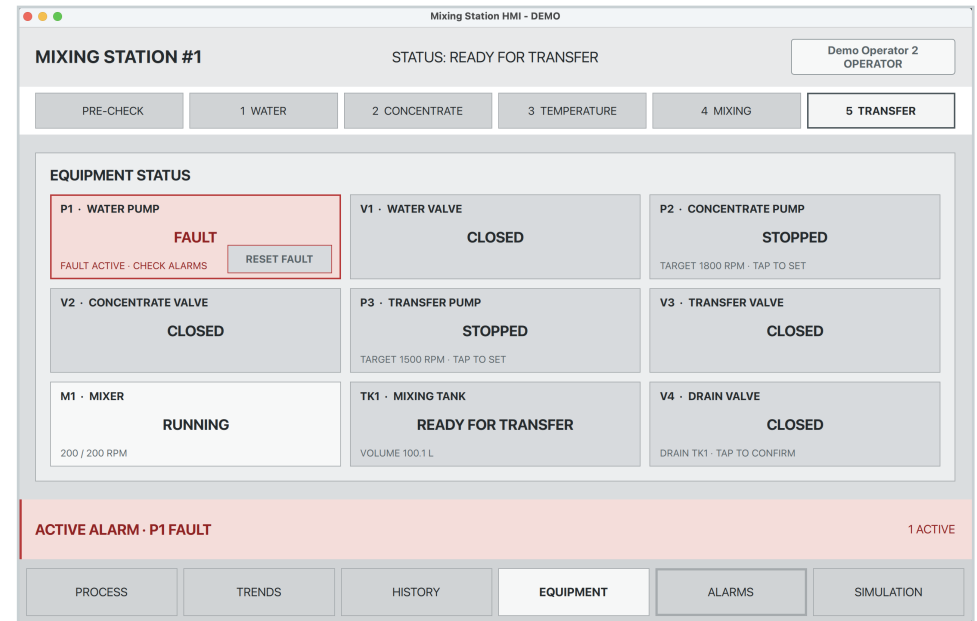


Fault visibility and recovery

Faults remain prominent in the process view and are actionable from their equipment cards.



A P1 fault pauses the batch and raises a visible alarm.



Equipment view exposes the affected device and fault reset.

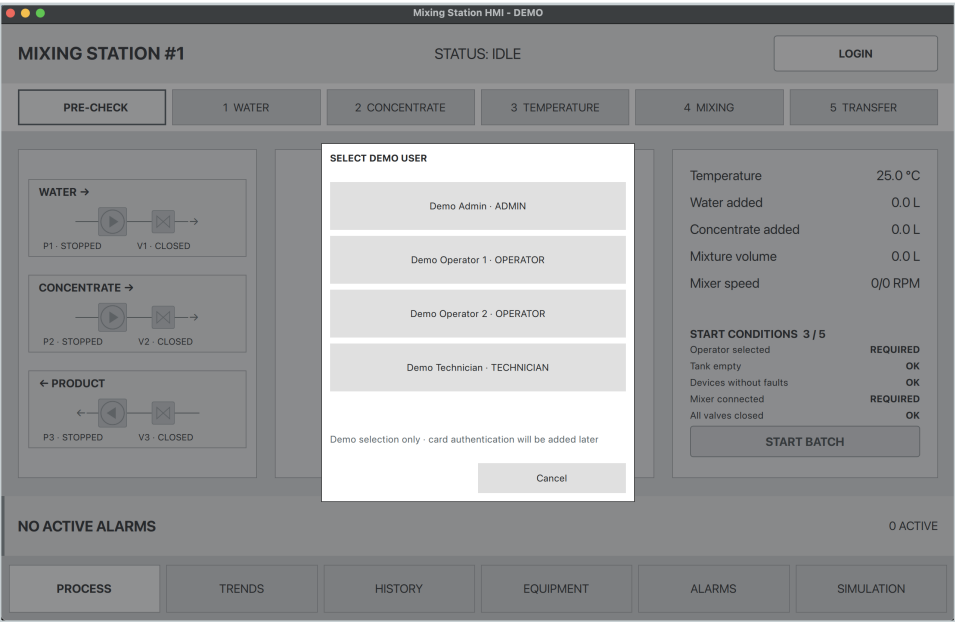
The Equipment reset is available to technician and admin roles. For a portfolio demonstration, an operator running a batch can use an explicitly labelled DEMO RESET in Simulation for P1 or M1. This override is recorded as a warning event.

Simulation behaviour

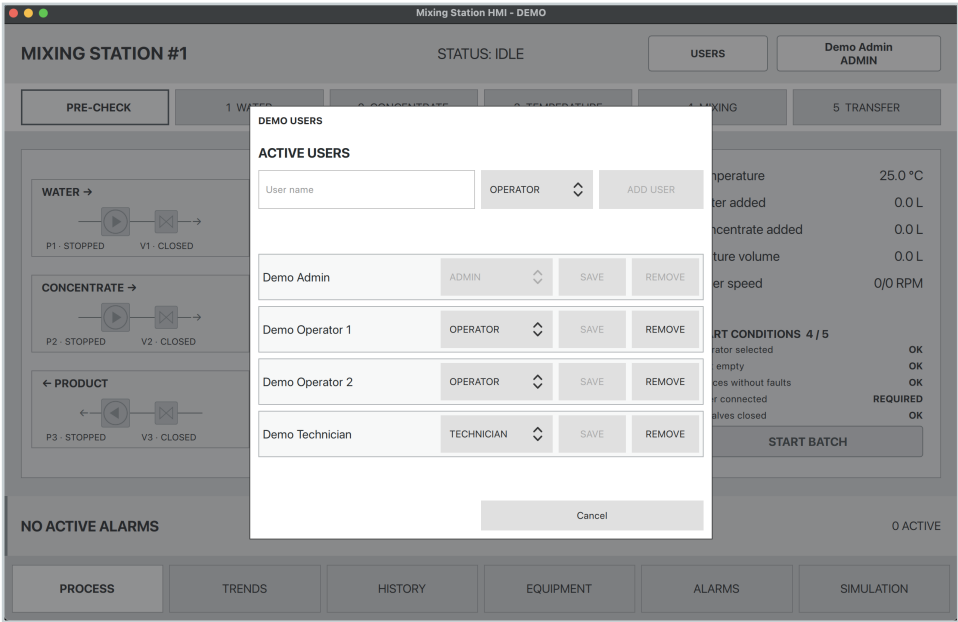
P1 and M1 fault injection/reset are implemented in this version. The interface must not be interpreted as a real safety or control system.

Demo roles and user management

The demo uses a database-backed user list, with permissions enforced by C++ services.



Demo login selects a user and role; it does not authenticate.



Admin view supports adding users and changing their roles/status.

OPERATOR: batch control

TECHNICIAN: + reset and drain

ADMIN: + user management

Removing a user from the active list leaves the database record in place so existing run and event references remain meaningful. Drain approval belongs to technician/admin; a future authentication step is indicated but not implemented.

Implementation and scope

How the current version is organised, built and bounded.

Interface and process

QML renders the 1280 x 800 screen. C++ device objects, BatchController and ProcessSimulator provide state, transitions and simulated values.

Persistence

SQLite schema version 1 contains users, runs, events and trend_samples. Repositories and list models supply History and user selection.

Testing

Qt Test targets cover equipment, batch transitions, roles, services and persistence. Run the project test suite with CTest.

Portfolio simulation boundary

No hardware I/O, physical interlocks, card/password authentication or production safety validation are included. The role selection and DEMO RESET exist to demonstrate UI and service behaviour.

Build

Requires Qt 6.8, Qt Quick, Qt SQL, Qt Test, CMake 3.16+ and a C++ compiler. Open CMakeLists.txt in Qt Creator or run:

```
cmake -S . -B build/portfolio
cmake --build build/portfolio
ctest --test-dir build/portfolio --output-on-failure
```

Database location

Created in the application data directory; the exact path is printed when the app starts.